Fiscal Year:	FY 2015	**Task Last Updated:**	FY 10/06/2016
PI Name:	Miller, Fletcher J Ph.D.		
Project Title:	Developing the Narrow Channel Apparatus as a NASA Standard Material Flammability Test		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	619-594-5791
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PI Web Page:			
City:	San Diego	**State:**	CA
Zip Code:	92182-1323	**Congressional District:**	53
Comments:	Previously at National Center for Space Exploration Research, NASA Glenn Research Center, until 2007.		
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2010	**End Date:**	12/31/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	6	**No. of Bachelor's Degrees:**	10
No. of Bachelor's Candidates:	10	**Monitoring Center:**	NASA GRC
Contact Monitor:	Olson, Sandra	**Contact Phone:**	216-433-2859
Contact Email:	Sandra.Olson@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date is now 12/31/2014 per NSSC information (Ed., 1/22/14) NOTE: End date is now 12/31/2013 per PI (Ed., 10/11/2012)		
Key Personnel Changes/Previous PI:	During the summer of 2010 Prof. George Sidebotham from Cooper Union participated in the research as a NASA Summer Faculty Fellow at NASA Glenn Research Center.		
COI Name (Institution):	Olson, Sandra (NASA Glenn Research Center) Wichman, Indrek (Michigan State University)		
Grant/Contract No.:	NNX10AD96A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This research will study flame spread over flammable solid materials such as plastic, paper, cloth, and hair, using a new apparatus designed to simulate microgravity flow conditions on Earth. We designed and built two new Narrow Channel Apparatuses at San Diego State, and one at Michigan State, that control environmental variables such as flow velocity, gas pressure, and oxygen concentration to reproduce space craft atmospheres. Additionally, the Narrow Channel suppresses buoyant flow by having a closely spaced ceiling on the channel, so that flow conditions closely resemble those found on space craft. The objective of the research is to prove the Narrow Channel Apparatus as a viable test method for opposed flow material flammability testing. This will allow NASA to screen materials in more realistic conditions than does the current NASA test which makes no account of microgravity. The experimental research consists of tests wherein we ignite various solids in the channels and measure how fast the flame spreads in various conditions. To accompany the experiments, we are developing a numerical model so that we can determine certain quantities such as heat loss to the channel ceiling that are difficult to measure. Note that this work continues under NASA Cooperative Agreement NNX14AC70A, “Developing the Narrow Channel Apparatus as a NASA Standard Material Flammability Test,” which began in January 2014 with Dr. Fletcher Miller continuing as Principal Investigator.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research encompasses flame spread and material flammability in narrow channels or gaps. We use this geometry to simulate microgravity conditions. However, many fire situations involve flames in narrow gaps. TWA Flight 800, which crashed off the coast of Canada, was brought down by a small fire spreading through some narrow gaps in the walls and ceiling near the cockpit, for instance. Many building fires involve fire spread inside walls. Fires in computers likewise can occur between the printed circuit boards. Thus, information gained from experiments about how flames spread in various atmospheric conditions in narrow gaps is useful in areas well beyond spacecraft fire safety. The numerical model we are developing will be validated against our experiments and can also be used to study flame spread in these gaps.
Task Progress:	[Ed. note October 2016: compiled from PI's final technical report] This research objective was to examine the flame spread in a Couette channel and determine if it’s an adequate apparatus to simulate microgravity in the study of flame spread. This research was divided into two sections, the first dealing with cold flow in pure Couette channel, and the second covering combustion in a linear velocity profile. Data for the first section is summarized below. It was apparent that a realistic finite length channel does not present us with a fully developed laminar flow profile that is exactly linear but is in fact slightly curved. This agrees well with other research done using Couette channels. The cause of this is the pressure differential along the entrance of the channel created by the top moving plate suction force on the fluid outside the channel as well as the pressure along the exit of the channel created by the flow leaving. The curviness of the profile is a function of Reynolds number and the length/height ratio of the channel. The pattern observed and the effect of the channel’s length, plate velocity and gap height was examined. As the channel gets longer the velocity profile becomes more linear and less curved, while the increase in the height of the channel reduces the linearity of the u-velocity profile. For the one moving top plate case, the higher the plate velocity the more curved the fully developed u-velocity profile becomes. The two moving plate, the u-velocity profile acts in a manner similar to a Hagen-Poiseuille channel with an adverse pressure differential along the channel. This matched the pattern expected from the analytical fully developed laminar u-velocity profile that was also derived during this research. The h, L and U0 were varied in such a manner to keep the h/L ratio and Re the same, the u-velocity and the pressure, did not change, which is what was expected. Only one ratio and one Reynolds number was utilized due to time restrictions. The entrance length was also examined and a relationship between it and the Reynolds number was determined for both the case at which only the top plate is moving and the case at which both top and bottom plates are moving at the same velocity. It was compared to the HP entrance length to find that. The second part of this study was the flame spread in a Couette channel. The channel utilized by Hung was the baseline of the model. The gap height was kept constant while varying the top plate velocity which showed a pattern that is similar for both 1g and 0g as well as similar to the experimental results. For the larger gap heights utilized, the flame at 0g was longer and higher than that of the 1g. This is due to the fact that in the 0g flame, the fuel pyrolyzed in front of the flame remains close to the surface allowing the 0g to spread faster before burning all the fuel behind it, which will diffuse upward until it meets the oxygen and combusts creating a longer and higher flame. However, increasing the height will reduce the oxygen gradient and hence slows down and shrinks the flame at 0g while for the 1g flame buoyancy takes over and the flame spread levels off and the flame height increases. Note that this work continues under NASA Cooperative Agreement NNX14AC70A, “Developing the Narrow Channel Apparatus as a NASA Standard Material Flammability Test,” which began in January 2014 with Dr. Fletcher Miller continuing as Principal Investigator.
Bibliography Type:	Description: (Last Updated: 06/15/2021)
Articles in Peer-reviewed Journals	Wichman IS, Olson SL, Miller FJ, Tanaya SA. "Experimental evaluation of flame and flamelet spread over cellulosic materials using the narrow channel apparatus." Fire and Materials. 2013 Nov;37(7):503-19. http://dx.doi.org/10.1002/fam.2143 , Nov-2013
Dissertations and Theses	Bornand G. "Polymethylmethacrylate Combustion In A Narrow Channel Apparatus Simulating A Microgravity Environment." Master’s Thesis, San Diego State University, Department of Mechanical Engineering, December 2014. , Dec-2014
Dissertations and Theses	Hamdan G. "Computational Modeling Of Flow And Combustion In A Couette Channel Simulating Microgravity." Master’s Thesis, San Diego State University, Department of Mechanical Engineering, December 2013. , Dec-2013